

INVESTMENTS

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CHAPTER 5

Risk, Return, and the Historical Record

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

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Study Guide · Prepared for Personal Use

LO1 · Interest Rate Determinants

Three basic forces determine the real interest rate: the **supply** of savings from households, the **demand** for funds from businesses wanting to finance investment, and net **government** actions (fiscal and monetary policy). While many different interest rates exist economy-wide, they tend to move together, so economists often speak of a single representative rate.

Real vs. Nominal Rates — The Fisher Equation

The **nominal interest rate** is the growth rate of your money; the **real interest rate** is the growth rate of your purchasing power. Because future inflation is uncertain, the real rate on an investment promising a fixed nominal rate is itself risky. The exact Fisher relation links the two; a simple approximation is widely used in practice (see Formulas page).

Taxes are levied on **nominal** income, not real income — so inflation erodes the after-tax real rate of return by more than the pre-tax real rate, since investors are taxed on the portion of nominal return that merely compensates for inflation, not just on true income.

LO2 · Comparing Rates of Return for Different Holding Periods

To compare returns across different maturities, we compute the total (holding-period) return earned over the full investment horizon from a zero-coupon bond's price, then convert to an annualized rate for comparison across horizons of different lengths (see Formulas page). Longer-horizon bonds often show a different **annualized** rate than shorter ones — the study of that pattern is the **term structure of interest rates**, covered fully in Chapter 14.

A distinct compounding convention, the effective annual rate (EAR), restates any return as if compounding occurred exactly once a year, allowing direct comparison of investments with different compounding frequencies (monthly, quarterly, continuous).

LO3 · Bills and Inflation, 1957–2016

Examining the historical record of Canadian T-bill rates, inflation, and the resulting real rate reveals a striking pattern: average interest rates were **higher** in the earlier part of the sample (1957–1986, averaging 7.21%) than in the more recent period (1987–2016, averaging 4.50%). The main driver was **inflation**, which averaged 5.26% in the earlier period versus just 2.27% more recently.

This comparison illustrates that the Fisher relation — the idea that nominal rates move to compensate for expected inflation — works far better when inflation itself is stable and predictable. When inflation is volatile and hard to forecast, investors cannot as reliably build the correct inflation compensation into the nominal rate they require, so realized real rates become more volatile too.

Practical takeaway: don't assume the historical average nominal T-bill rate is a stable predictor of future real returns — the underlying inflation regime matters enormously, and that regime can shift for extended periods.

LO4 · Risk and Risk Premiums

Holding-Period Returns and Scenario Analysis

The realized return on a risky asset over a holding period depends on the ending price and any cash distributions received. Because the future is uncertain, we describe a risky investment using a **scenario analysis**: a set of possible outcomes, each with an assigned probability. From this, we compute the **expected return** (the probability-weighted average of possible returns) and the **variance / standard deviation** (the probability-weighted measure of dispersion around that expected return — our measure of risk).

The Risk Premium and Risk Aversion

The **risk premium** is the expected return on a risky asset in excess of the risk-free rate — compensation investors demand for bearing risk. The degree to which investors demand this compensation reflects their **risk aversion**. A risk-averse investor rejects a fair gamble (one with an expected value of zero) because the pain of a loss looms larger than the pleasure of an equal-sized gain — a consequence of the diminishing marginal utility of wealth. This concept is developed fully in Chapter 6, where it is used to derive optimal portfolio choice.

LO5 · Time Series Analysis of Past Rates of Return

A **scenario analysis** is forward-looking: we specify probabilities and payoffs before the fact. A **time series** of historical returns is different — we observe only realized dates and holding-period returns, with no explicit probabilities attached. We must therefore **infer** the underlying probability distribution from the pattern of past outcomes, typically by treating each historical period as one equally likely draw from that distribution.

Arithmetic vs. Geometric Average Returns

The **arithmetic average** (simple mean of period-by-period returns) provides an **unbiased estimator of expected future one-period return** — the right measure to use when forecasting next period's return. The **geometric average** (the constant per-period return that would compound to the same total, multi-period return actually earned) is the correct measure of **actual historical performance** over the full sample period.

The geometric average is always *less than or equal to* the arithmetic average, with the gap growing as volatility increases — for normally distributed returns, the expected gap is approximately half the variance of returns. This asymmetry arises because a large negative return has a proportionally bigger drag on compounded wealth than an equally large positive return has a boost, and using the arithmetic average corrects for that downward bias when estimating expected future performance.

LO6 · The Normal Distribution

The bell-shaped **normal distribution** appears throughout finance because variables driven by many small, independent random influences tend toward normality. If asset returns are normally distributed, the mean and standard deviation together provide a **complete** description of risk — which is why the Sharpe ratio (see Formulas page) is a sufficient measure of risk-adjusted performance under this assumption.

Deviations from Normality — Higher Moments

Real-world returns often deviate from normality. Statisticians describe distributions using **moments**: the 1st moment is the mean, the 2nd is the variance. Two higher moments matter for risk management:

- **Skewness (3rd moment)** — measures asymmetry. Negative skew means the distribution has a longer/fatter left (loss) tail than a normal distribution would predict — more risk of a large loss than the standard deviation alone suggests.
- **Kurtosis (4th moment)** — measures the degree of "fat tails." Higher kurtosis means both extreme gains and extreme losses occur more often than a normal distribution predicts.

Alternative Risk Measures for Non-Normal Returns

Measure	Definition
Value at Risk (VaR)	The loss at a low percentile of the return distribution — e.g., the 1% VaR is the cutoff separating the worst 1% of outcomes from the rest. For normally distributed returns: $\text{VaR}(1\%) = \text{Mean} - 2.33 \times \text{SD}$ (see Formulas page).
Expected Shortfall (ES)	Also called conditional tail expectation. The average loss given that you are already in the worst-case tail (e.g., the average of the worst 1% of observations) — more informative than VaR because it captures the severity of extreme losses, not just the cutoff.
Lower Partial Standard Deviation (LPSD)	Computed like ordinary standard deviation, but using only negative deviations from the risk-free rate (not all deviations from the mean) — focuses risk measurement specifically on downside outcomes. Underlies the Sortino ratio, a downside-risk analogue to the Sharpe ratio.

LO7 · Historic Returns on Risky Portfolios

Applying these tools to history: comparing Treasury bills, Treasury bonds, and a broad, value-weighted stock market index (all NYSE/AMEX/NASDAQ-listed stocks, 1926–2015) confirms the risk–return hierarchy predicted by theory.

Asset Class	Standard Deviation	Excess Return over T-bills (Risk Premium)
Treasury Bills	Lowest (~ known nominal return each period)	— (the base rate)
Treasury Bonds	10.02%	Lower than stocks
Common Stocks (broad index)	20.28% (~double T-bonds)	8.30% per year

Stocks' standard deviation (20.28%) is roughly double that of T-bonds (10.02%) and many times that of T-bills — and that greater risk is compensated by a substantially higher average return, an 8.30% annual risk premium over the full sample. The frequency distribution of returns also shows evidence of **fat tails**: actual extreme returns (both very high and very low) occur more often than a normal distribution with the same mean and variance would predict, particularly evident during periods like the Great Depression.

Why this matters going forward: this long-run historical risk premium (roughly 8% for large stocks over T-bills) is commonly used as a starting benchmark for forecasting expected future equity returns — while recognizing it is an average over a long, volatile sample, not a guarantee for any given year.

Chapter 5 — Big-Picture Takeaways

- Real returns matter more than nominal returns; the Fisher equation links the two, and taxes bite into nominal (not real) income, magnifying inflation's bite on after-tax real returns.
- Historical interest rate levels are heavily driven by the inflation regime of the period — rates were higher and more volatile when inflation itself was higher and less predictable.
- Investors demand a risk premium to hold risky assets; the size of that premium reflects the market's aggregate risk aversion.
- Use the arithmetic average to forecast next period's expected return; use the geometric average to describe actual realized performance over a past multi-period sample.
- Standard deviation is a complete risk measure only under normality; skewness, kurtosis, VaR, ES, and LPSD are needed when real-world fat tails and asymmetry matter.
- The long-run historical record confirms the risk–return trade-off: stocks show roughly double the volatility of bonds and are compensated with a substantially higher average return.

Chapter 5 is formula-dense — these are the core relationships tested throughout the rest of the course.

F1 | Fisher Equation — Exact Real Rate

$$1 + r_{\text{real}} = (1 + r_{\text{nom}}) / (1 + i)$$

$$\text{Approximation: } r_{\text{real}} \approx r_{\text{nom}} - i$$

r_{nom} = nominal rate; *i* = inflation rate. The approximation overstates the real rate by a factor of $(1 + i)$; more accurate for small *i*.

F2 | Annualized Return from a Zero-Coupon Price

$$\text{Total return over horizon } T: r(T) = [100 / P(T)] - 1$$

$$\text{Annualized (EAR-style): } 1 + r_{\text{annual}} = [1 + r(T)]^{(1/T)}$$

P(T) = price of a \$100-face zero-coupon bond maturing in *T* years. Lets you compare returns across different investment horizons on an apples-to-apples annual basis.

F3 | Expected Return & Variance (Scenario Analysis)

$$E(r) = \sum [p(s) \times r(s)]$$

$$\text{Var}(r) = \sum [p(s) \times (r(s) - E(r))^2]$$

p(s) = probability of scenario *s*; *r(s)* = return in scenario *s*. Standard deviation $SD = \sqrt{\text{Var}(r)}$.

F4 | Geometric (Time-Weighted) Average Return

$$1 + g = [(1+r_1)(1+r_2)\dots(1+r_n)]^{(1/n)}$$

The single constant annual return that compounds to the same ending wealth as the actual sequence of historical returns. Always $\leq$ arithmetic average; the gap widens with volatility ($\approx$ half the variance for normal returns).

F5 | Sharpe (Reward-to-Volatility) Ratio

$$\text{Sharpe Ratio} = \text{Risk Premium} / \text{SD of Excess Returns}$$

$$= [E(r_p) - r_f] / \sigma_p$$

Measures reward per unit of risk using excess (not total) returns. Widely used to evaluate manager performance, though it is inadequate for individual (non-diversified) assets.

F6 | Value at Risk (Normal Distribution)

$$\text{VaR}(1\%, \text{normal}) = \text{Mean} - 2.33 \times \text{SD}$$

-2.33 is the 1st percentile of the standard normal distribution. VaR marks the cutoff separating the worst 1% of outcomes from the rest; for non-normal (fat-tailed) returns, empirical VaR is estimated directly from sorted historical observations instead.

EXAMPLE 5.1 | Approximating the Real Rate

Given: A 1-year CD offers a nominal rate of 8%. You expect inflation of 5% over the coming year.

Approximation formula:

$$r_{\text{real}} \approx 8\% - 5\% = 3\%$$

Exact Fisher formula:

$$1 + r_{\text{real}} = 1.08 / 1.05 = 1.0286 \rightarrow r_{\text{real}} = 2.86\%$$

The approximation overstates the real rate by 0.14 percentage points (14 basis points). The gap grows with the inflation rate and shrinks toward zero for very low inflation.

EXAMPLE 5.9 | Computing the Sharpe Ratio

Given: A scenario analysis for a proposed stock-index fund investment produces a risk premium of 5.76% and a standard deviation of excess returns of 19.49%.

$$\text{Sharpe Ratio} = 5.76 / 19.49 = 0.30$$

A Sharpe ratio of 0.30 is roughly in line with the long-run historical performance of stock-index funds. The Sharpe ratio is an adequate risk-adjusted performance measure for diversified portfolios, but inadequate for evaluating individual stocks, since it ignores diversifiable risk that a rational investor would not hold unhedged.

PRACTICE PROBLEM | Foothills Wealth Advisors**Scenario**

You are an analyst at Foothills Wealth Advisors in Calgary, preparing client materials on historical risk and return.

Part A — Real vs. Nominal Rates

A client's GIC offers a nominal rate of 6%. Expected inflation over the term is 3.5%.

- (i) Calculate the approximate real rate using the simple formula.
- (ii) Calculate the exact real rate using the Fisher equation.
- (iii) By how many basis points does the approximation overstate the exact real rate?

Part B — Scenario Analysis

A proposed equity position has three possible outcomes: Boom (probability 0.25, return 30%), Normal (probability 0.50, return 12%), Recession (probability 0.25, return -15%).

- (i) Calculate the expected return, $E(r)$.
- (ii) Calculate the variance and standard deviation of returns.
- (iii) If the risk-free rate is 4%, calculate the risk premium on this position.

Part C — Arithmetic vs. Geometric Average

A fund produced the following annual returns over 4 years: Year 1: +25%, Year 2: -10%, Year 3: +18%, Year 4: -5%.

- (i) Calculate the arithmetic average annual return.
- (ii) Calculate the geometric average annual return.
- (iii) Which measure should Foothills use to forecast next year's expected return, and which should it use to report the fund's historical 4-year performance? Explain the distinction.

Part D — Sharpe Ratio & Risk Measures

A client portfolio has a risk premium of 6.40% and a standard deviation of excess returns of 16.00%. The risk-free rate is 4.5%.

- (i) Calculate the portfolio's Sharpe ratio.
- (ii) If the portfolio's return distribution is approximately normal with mean 10.9% and SD 16.00%, calculate the 1% VaR.
- (iii) A colleague suggests the portfolio's returns show negative skew and high kurtosis. Explain in plain language what this means for the client's actual risk, beyond what the Sharpe ratio alone captures.

Hints & Guidance

Part A tests LO1 and Formula F1 (mirror Example 5.1). Part B tests LO4 and Formula F3. Part C tests LO5 and Formula F4 — remember, geometric average uses the product of $(1+r)$ terms, not a simple sum. Part D tests LO6–LO7 and Formulas F5–F6 (mirror Example 5.9 for (i)).